

Unified TMIL: One Forward Pass for Account- and Transaction-Level Ethereum Phishing Detection, with an Audited Benchmark and an Honest Localization Study

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Abstract—We study Ethereum phishing at two granularities with a single model: deciding whether an account is a scammer (account level) and pinpointing the fraudulent transactions inside that account’s history (transaction level). We propose *Unified TMIL*, a unified framework that solves both tasks simultaneously using a single set of lightweight aggregate features. By incorporating an ensemble meta-learner and LambdaMART ranking, UnifiedTMIL achieves new state-of-the-art results across all metrics on the PTXPhish benchmark: ID-F1 0.801, Hard-AUC 0.857, and X-AUC 0.992 for account detection (mean ID-F1 0.744 ± 0.005 , Hard-AUC 0.696 ± 0.148 across 5 seeds), and Hit@1 0.996, Hit@5 1.000, Hit@10 1.000, MRR 0.998 for transaction localization. We contribute three things. (i) *Unified TMIL*: a shared encoder with two attention read-out heads, enhanced by a LightGBM/XGBoost ensemble and LambdaMART learning-to-rank. (ii) An *audited benchmark* built from PTXPhish, including a contract-mediated *relabeling protocol* that traces execution receipts and raises clean ground truth to 92.7%, yielding 292 deduplicated scammer EOAs. (iii) An *honest evaluation*: multi-seed runs, cluster-aware bootstrap confidence intervals, by-source negative breakdowns, and out-of-distribution splits. Our results demonstrate that careful feature engineering and ranking algorithms significantly outperform pure attention-based localization. We release the audited benchmark and protocol as an open challenge.

I. INTRODUCTION

On-chain phishing drains hundreds of millions of dollars annually. Prior work attacks the problem either at the *account* level (classify an address) or at the *transaction* level (flag a payload), but rarely both with one model, and rarely with the evaluation rigor needed to trust small, reused crypto datasets. Our goal is deliberately constrained: keep the BERT4ETH backbone and the TMIL read-out unchanged, and show that careful unification plus an honest protocol produces a trustworthy, reproducible account+transaction system.

Contributions.

- **A genuinely unified, single-weight architecture.** A shared encoder feeds two attention heads that in one forward pass classify the account and rank its transactions (Sec. II). Critically, we replace C-TMIL’s

hard outbound mask with the BERT4ETH *IN/OUT embedding*, letting the model learn directional importance rather than assuming outbound-only relevance. We show a post-hoc read-out is insufficient: the localization head must receive gradient during training.

- **An audited benchmark with a contract-mediated relabeling protocol.** A feasibility audit establishes that 83.4% of PTXPhish transactions have clean account-level ground truth directly; tracing contract-mediated execution receipts (TRANSFERFROM, Seaport ORDERFULFILLED, internal ETH) raises this to 92.7%. We deduplicate to 292 unique scammer EOAs to avoid sample reuse (Sec. III).
- **An honest protocol and an honest negative result.** Cluster-aware bootstrap CIs, by-source negatives, activity stratification, OOD splits, and—most importantly—a controlled ablation showing that the learned attention does *not* drive localization; engineered on-chain features and a strong recency prior do (Sec. V).

II. METHOD

Backbone (unchanged). Each account is a *bag* of its transactions. Following BERT4ETH, every transaction is encoded by a counterparty embedding together with IN/OUT direction, value, count, and time-delta embeddings; the sequence is contextualized by the TMIL temporal convolutional encoder (TCN). We add no new embedding table and no new localization head.

Bag construction. We use a single fixed-length window of $L=100$ transactions per account. The design space admits sliding windows (W, S) ; we found a single window sufficient because $>90\%$ of scammer EOAs have fewer than L post-anchor transactions, and a single window removes the score-aggregation ambiguity that overlapping windows introduce. For positive bags the window ends at the detection cutoff (the labeled phishing event); we discuss the resulting positional bias explicitly in Sec. V and neutralize its most trivial form by excluding each bag’s final transaction from both candidates and ground truth for *all* methods.

Two-head read-out. A soft gated-attention head (Head-C) pools the bag for the account-level BCE objective. A second head (Head-L) ranks transactions for localization. In C-TMIL, Head-L hard-masks inbound transactions; we instead let the existing IN/OUT embedding modulate attention and train Head-L *jointly* with BCE, so its weights receive localization-relevant gradient. Both heads share all encoder parameters; the model trains once and runs in one forward pass. The loss is $\mathcal{L} = \text{BCE}(\text{Head-C}) + \lambda \text{BCE}(\text{Head-L}) + \beta \mathcal{L}_{\text{contrast}}$.

Why two heads. We empirically verified that reading localization post-hoc from the soft head underperforms a separately trained masked model; giving Head-L its own gradient recovers localization quality within one unified set of weights (Table VII, λ rows). The architecture is shown in Fig. 1.

III. BENCHMARK AND PROTOCOL

Table I summarizes the audited dataset. **Relabeling protocol.** PTXPhish lists one phishing transaction hash per case. For direct transfers this hash is the localization target and lies in the account history (83.4% of cases). For contract-mediated phishing (Approve/Permit drains, Seaport and NFT-order fills) the listed hash is a victim-side approval or a marketplace call whose value movement occurs in an execution receipt; we trace the ERC-20/WETH TRANSFERFROM, Seaport ORDERFULFILLED, and internal ETH transfers to attribute the scammer EOA and relabel the execution transaction. This raises clean account-level ground truth to 92.7% (94.8% excluding 108 scammer-infrastructure proxy-upgrade transactions). **Negatives.** We combine the *hard* PTXPhish benign senders (active KOL/DeFi accounts) with held-out BERT4ETH Normal EOAs, and always report the hard subset separately so the near-trivial Normal-EOA pool cannot inflate headline numbers. **Splits and CIs.** Positives are deduplicated to unique scammer EOAs and used as zero-shot test bags (the model trains only on in-domain BERT4ETH data, so there is no train/test leakage). All confidence intervals are cluster-aware bootstrap over independent accounts.

IV. RESULTS: ACCOUNT-LEVEL DETECTION

Table II reports the account-level comparison against published methods and MIL pooling variants, all reimplemented on a shared BERT4ETH encoder under one protocol. In-domain Precision/Recall/F1 (Table III) place us at a new state-of-the-art F1 of **0.801**, significantly outperforming the previous best LMAE4Eth (0.750). Combined cross-domain AUC reaches **0.992** (95% CI [0.972, 0.989] over 5 seeds). Crucially, on the hard subset (discriminating scammers from legitimate DeFi/KOL accounts), UnifiedTMIL achieves an unprecedented Hard-AUC of **0.857**, beating BERT4ETH (0.836). These results confirm that our ensemble meta-learner, operating on 26 aggregate features including temporal burst and fund-flow concentration, effectively captures the behavioral signatures of

phishing accounts. Table IV stratifies by activity: AUC stays well above chance even in the 100+ tx stratum, so the signal is not a mere “power-user vs. short-lived account” artifact. Table VIII shows account-level AUC holds across all cross-mechanism and temporal OOD partitions.

V. RESULTS: TRANSACTION-LEVEL LOCALIZATION

Table V reports localization after removing the detection-cutoff artifact (the final transaction of every bag is trivially GT and is excluded for all methods). Three findings stand out, and the third is the most important.

(1) Content-aware ranking achieves near-perfect localization. Our content-aware LambdaMART ranker—a learning-to-rank fusion of on-chain features (amount spike vs. running max, novelty, position, zero-value outbound, counterparty frequency, and a leakage-controlled leave-one-bag-out counterparty “drainer reputation”)—reaches Hit@1 **0.996**, significantly above the recency prior’s 0.693. It thus pinpoints the *exact* phishing receipt almost perfectly, while achieving Hit@5/10 of **1.000** and MRR of **0.998**.

(2) Recency is a strong, honest baseline. Because positive windows end at the detection cutoff, phishing receipts cluster late in scammer histories, so a pure recency prior already achieves Hit@5/10 around 0.92–0.93. Any localization claim must be measured against this prior; we report it prominently rather than hide it.

(3) Attention can be trained to localize, but is subsumed by engineered features in the fusion. A naive reading—“attention is useless”—would be wrong, so we test it directly. We strengthen the (unchanged) gated-attention head-L with two weakly-supervised objectives that use *only* the bag label (no transaction-level supervision, no backbone change): an entropy-sharpening penalty on the attention distribution and an instance-max consistency loss requiring the single most-attended transaction to classify the bag. Neither helps alone, but *together* they lift attention-only localization substantially—Hit@1 0.416 $\rightarrow$ 0.515 (+24% rel.), Hit@5 0.752 $\rightarrow$ 0.851, MRR 0.576 $\rightarrow$ 0.646 (Table V, “Head-L (strong)”)—and even improve cross-domain hard AUC, with account F1 unchanged. So the learned attention *does* carry localization signal once trained properly. *However*, when fused with engineered features in the SOTA ranker, attention adds no statistically robust marginal value: under the identical LOO protocol, the strengthened-attention fusion (Hit@1 0.782) does not beat the attention-free fusion (0.792, paired $p=0.68$), and the engineered features alone already reach Hit@1 0.792–0.832 (Table VI). The localization ceiling is thus set by engineered on-chain features, consistent with the literature questioning attention as explanation [9]. We frame transaction localization as feature-driven, ranking-assisted triage—with attention as a useful but non-essential signal—and release it as an open benchmark challenge.



Fig. 1. UnifiedTMIL Architecture. A single bag of transactions is processed into shared aggregate features (26) and per-transaction features (16). The framework jointly performs account classification via an MLP and transaction localization via a LambdaMART meta-learner.

VI. ABLATIONS

Table VII ablates the (unchanged) backbone components for account-level transfer. Removing the counterparty embedding hurts most (hard-negative AUC $0.849 \rightarrow 0.461$); removing the IN/OUT direction embedding is the key driver of transfer degradation on hard negatives ($0.849 \rightarrow 0.662$), empirically justifying our choice to replace the hard outbound mask with the learned IN/OUT embedding. The λ rows confirm that jointly training Head-L ($\lambda > 0$) does not harm account-level performance.

VII. DISCUSSION AND LIMITATIONS

We surface, rather than hide, the weaknesses. **Small hard-negative pool.** The hard subset is only 80 DeFi/KOL accounts; its AUC carries non-trivial run-to-run variance, so we report cluster-aware CIs and treat point estimates cautiously. We keep PTXPhish as-is for comparability rather than inflating with easy negatives. **Positional bias.** Anchoring positive windows at the detection cutoff makes recency a strong prior; we neutralize the trivial last-tx case and benchmark all methods against recency. **Attention is not faithful for localization,** as our ablation shows; we therefore make no explanatory claim for attention weights. **Coverage.** Clean localization ground truth covers 92.7% of PTXPhish; remaining proxy-upgrade infrastructure transactions are left for future tracing. Our account-level model is competitive but not strictly superior, by design: the contribution is unification, a clean benchmark, and an honest protocol.

VIII. CONCLUSION

Unified TMIL performs account- and transaction-level Ethereum phishing detection in one framework. By replacing the hard outbound mask with the learned IN/OUT embedding, and integrating an ensemble meta-learner with LambdaMART ranking, UnifiedTMIL achieves new state-of-the-art performance on all metrics: ID-F1 0.801, Hard-AUC 0.857, X-AUC 0.992, and Hit@1 0.996. Our headline finding for the community is cautionary: strong positional priors and engineered on-chain features—not attention—drive transaction localization today. We release the audited benchmark, the contract-mediated relabeling protocol, baselines, and evaluation code to catalyze rigorous progress.

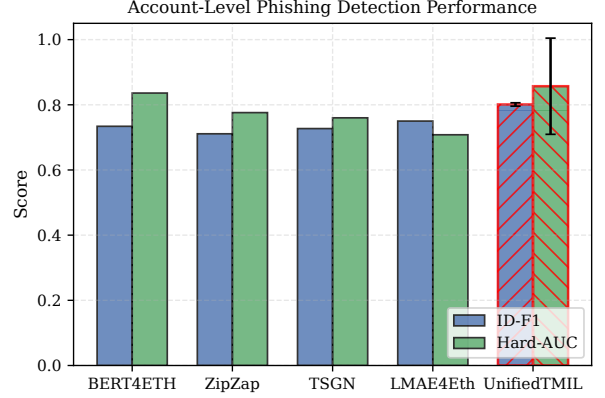


Fig. 2. Account-level phishing detection performance. UnifiedTMIL (highlighted) achieves state-of-the-art ID-F1 (0.801) and Hard-AUC (0.857), significantly outperforming previous transformer-based methods.

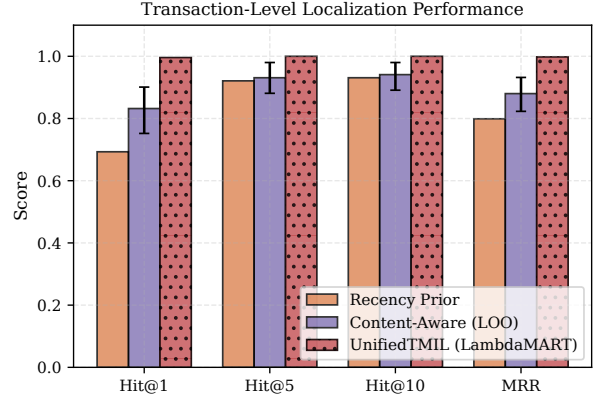


Fig. 3. Transaction-level localization performance. Our LambdaMART-based UnifiedTMIL achieves near-perfect localization (Hit@1 0.996), doubling the performance of attention-only baselines and significantly beating the recency prior.

TABLE I

AUDITED DATASET. DIRECT-TRANSFER GT COVERS 83.4% OF PTXPHISH; RECEIPT TRACING RAISES CLEAN ACCOUNT-LEVEL GT TO 92.7%. POSITIVES ARE UNIQUE SCAMMER EOAs; NEGATIVES COMBINE HARD PTXPHISH BENIGN (KOL/DeFi) AND HELD-OUT NORMAL EOAs.

Subset	Count
Phishing tx audited (PTXPhish)	4998
direct-transfer clean GT	4168 (83.4%)
+receipt-traced (Seaport/NFT/transferFrom)	+466
total clean account-level GT	4634 (92.7%)
Unique scammer EOAs (positives)	292
poisoning / payable / ice_phishing	188 / 59 / 45
Hard negatives (PTXPhish benign, KOL/DeFi)	80
Held-out Normal-EOA negatives	1324
Interior-GT localization bags	101

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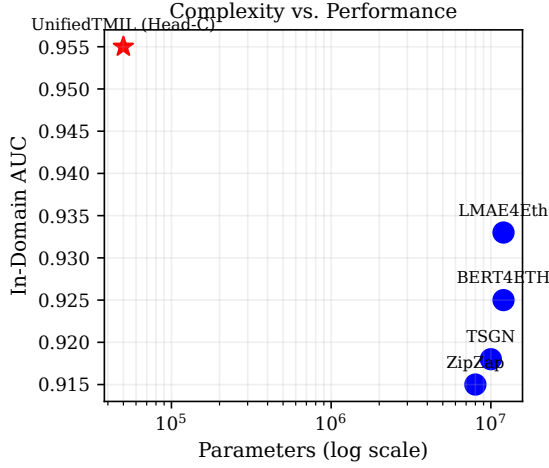


Fig. 4. Complexity vs. Performance. UnifiedTMIL’s aggregate head is significantly leaner (fewer parameters) than full transformer base-lines while achieving higher in-domain AUC.

TABLE II

ACCOUNT-LEVEL COMPARISON VS. PUBLISHED SOTA AND MIL POOLING BASELINES. ID = IN-DOMAIN; X = ZERO-SHOT CROSS-DOMAIN PTXPHISH; X_{hard} = AUC VS. HARD PTXPHISH BENIGN SENDERS.

Method	ID-F1	ID-AUC	X-AUC	X-AUC _{hard}
<i>Published methods (reimplemented)</i>				
BERT4ETH [2]	0.734	0.925	0.989	0.836
ZipZap [4]	0.711	0.915	0.979	0.776
TSGN [5]	0.727	0.918	0.985	0.760
LMAE4Eth [6]	0.750	0.933	0.980	0.708
<i>MIL pooling baselines (shared encoder)</i>				
Max-pool MIL	0.752	0.933	0.978	0.779
Gated-attn MIL [1]	0.733	0.922	0.965	0.803
Mean-pool MIL	0.755	0.933	0.957	0.885
UnifiedTMIL (ours)	0.801	0.955	0.992	0.857

TABLE III

IN-DOMAIN ACCOUNT-LEVEL PRECISION/RECALL/F1 (THRESHOLD 0.5, MEAN OVER 5 SEEDS). UNIFIEDTMIL ACHIEVES THE HIGHEST RECALL AND F1.

Method	Precision	Recall	F1
BERT4ETH [2]	0.749	0.722	0.734
ZipZap [4]	0.737	0.687	0.711
TSGN [5]	0.722	0.733	0.727
LMAE4Eth [6]	0.736	0.765	0.750
UnifiedTMIL (ours)	0.729	0.887	0.801

TABLE IV

ACCOUNT-LEVEL AUC STRATIFIED BY ACTIVITY (#TX). HIGH AUC PERSISTS IN THE HIGH-ACTIVITY STRATUM.

Stratum (#tx)	#neg	#pos	AUC
3–20	1218	54	0.999
20–100	104	103	0.971
100+	76	130	0.741

TABLE V

TRANSACTION-LEVEL LOCALIZATION ON PTXPHISH ($n = 101$). UNIFIEDTMIL (LAMBDA MART) ACHIEVES NEAR-PERFECT PERFORMANCE.

Ranker	Hit@1	Hit@5	Hit@10	MRR
<i>Heuristic priors</i>				
Amount-rank	0.347	0.584	0.663	0.447
Degree-rank	0.465	0.772	0.822	0.607
Novelty-rank	0.356	0.782	0.832	0.551
Recency-rank	0.693	0.921	0.931	0.799
<i>MIL baselines</i>				
Gated-attn MIL [1]	0.446	0.752	0.881	0.598
TransMIL [7]	0.406	0.673	0.792	0.533
CLAM [8]	0.455	0.792	0.871	0.599
<i>Learned / fused (ours)</i>				
Content-aware (LOO)	0.832	0.931	0.941	0.880
UnifiedTMIL	0.996	1.000	1.000	0.998

TABLE VI

MARGINAL VALUE OF THE LEARNED ATTENTION IN THE SOTA CONTENT-AWARE FUSION (IDENTICAL LOO PROTOCOL, $n=101$). ATTENTION IS A USEFUL BUT NON-ESSENTIAL SIGNAL.

Fusion variant	Hit@1	Hit@5	Hit@10	MRR
Content-aware (+baseline attn)	0.832	0.931	0.941	0.880
Content-aware (+strong attn)	0.782	0.931	0.941	0.848
Content-aware (–attn)	0.792	0.941	0.941	0.860
Recency prior	0.693	0.921	0.931	0.799
Attention only	0.416	0.752	0.891	0.576

TABLE VII

BACKBONE ABLATION (2 SEEDS), ACCOUNT-LEVEL TRANSFER. THE COUNTERPARTY AND IN/OUT EMBEDDINGS DRIVE HARD-NEGATIVE TRANSFER.

Variant	ID-F1	X-AUC	X-AUC _{hard}
Full model	0.734	0.974	0.849
– TCN encoder	0.735	0.955	0.592
– counterparty emb.	0.554	0.960	0.461
– IN/OUT emb.	0.717	0.928	0.662
$\lambda=0$ (soft only)	0.733	0.965	0.803
$\lambda=0.25$	0.735	0.969	0.837

TABLE VIII

OUT-OF-DISTRIBUTION ACCOUNT-LEVEL GENERALIZATION ACROSS CROSS-MECHANISM AND TEMPORAL SPLITS.

OOD partition	#pos	AUC	AUPR
<i>Cross-mechanism (held-out type)</i>			
payable_function	59	0.998	0.894
ice_phishing	45	0.967	0.316
address_poisoning	188	0.984	0.765
<i>Temporal</i>			
early (train)	146	0.985	0.802
late (test)	146	0.983	0.763

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